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MILITARY SPECIFICATION

THERMOMETERS, SELF-INDICATING, LIQUID-IN-GLASS, AIR AND WATER, STANDARD

(This specification has been approved by the Department of Defense, and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.)

1. SCOPE

1.1 Scope.- This specification covers wall mounting thermometers for determining the temperature of surrounding air, for outdoor or indoor use, and suspension type thermometers for determining air or water temperatures. These thermometers are non-mercurial and are not to be used for meteorological purposes.

1.2 Classification.- Thermometers shall be of the following types and classes as specified (see 6.2):

Type I - General purpose, air
  Class 1 - Without guards
  Class 2 - With wire mesh guards
  Class 3 - With plate guards
Type II - Shiphold suspension, air or water

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL
RR-C-440 - Cloth, Wire, Industrial.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Fiber.
PPP-B-665 - Boxes; Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676 - Boxes, Set-Up, Paperboard.
PPP-T-76 - Tape, Pressure-Sensitive Adhesive, Paper, Water Resistant.
PPP-T-97 - Tape; Pressure-Sensitive Adhesive, Filament Reinforced.

MILITARY
MIL-B-10377 - Boxes: Wood-Cleated, Veneer, Paper Overlaid.
MIL-E-15090 - Enamel, Equipment, Light-Gray (Formula No. 111).

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U.S. Military Property.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N.Y.)

3. REQUIREMENTS

3.1 General requirements.-

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3.1.1 Thermometers shall consist of a glass tube, bulb, scale and case as shown on figures 1 and 2.

3.1.2 Tube and bulb.-

3.1.2.1 Material.- The tube and bulb shall be of suitable glass, well seasoned. The tube shall be of the magnifying type, with a reading angle of not less than 10 degrees in a plane perpendicular to the tube, and shall be provided with a strip of white enamel at the back.

3.1.2.2 Dimensions.- The tube shall be of suitable cross section, 1/4 ± 1/32 inch in diameter, or approximately equivalent in any shape other than round. The bulb shall be cylindrical, approximately 1/5 inch in diameter by 7/8 inch long, or spherical, from 0.35 to 0.550 inch in diameter. The length of tube, including bulb, shall be not less than 10 inches nor more than 10-1/2 inches. A knob shall be incorporated at the top end of the tube to project through a hole in the scale to prevent slippage after assembly.

3.1.3 The tube and bulb shall contain red organic liquid free from air bubbles. Drainage and color stability characteristics shall be tested as specified in 4.4.2 and 4.4.3 respectively.

3.1.4 Range limits.- The scale shall be graduated to 2 degrees in units according to the Fahrenheit (F.) system, the lower limit range between minus 20 and 50 degrees and the upper limit ranging between plus 120 and 130 degrees. The minimum allowable length of the minimum range, minus 20 to plus 120 degrees, shall be 6-1/4 inches. The limit of error at any temperature on the scale shall not be greater than 2.0 degrees. The 10 degree marks shall be longer than the other marks and shall be figured on the scale in clear legible figures having a minimum height of 5/16 inch for type I thermometers and 1/8 inch for type II thermometers.

3.2 Specific requirements.-

3.2.1 Type I, general purpose, air.-

3.2.1.1 Scale.- The tube shall be mounted on a metal scale having a white enameled face and the scales shall be mounted on a case of a design similar to figure 1.

3.2.1.2 Case.- The case shall be made of brass or steel and shall include screw holes at each end for securing to a wall or bulkhead. The case shall be formed with flanges extending forward to a distance greater than the bulb for protection, and to provide a rigid housing for the glass tube. Unless otherwise specified (see 6.2), cases shall be treated and painted in accordance with the manufacturer's best commercial practice. When specified cases shall be coated with gray enamel conforming to type I or II, class 2 of Specification MIL-E-15090 or the scale may be integral with the case, and, where integral, the case may be the same color as the scale.

3.2.1.3 Attachment.- The tube shall be mounted to the scale by means of metal clamps secured by eyelets and the knob on the tube shall fit into a hole in the scale to prevent slippage. The scale shall be attached to the case and adjusted so as to leave a free circulation of air around the bulb, or, if the scale is integral with the case, the tube shall be mounted directly on the case by means of metal clamps secured by eyelets.

3.2.1.4 Graduations.- The graduation marks shall be neatly, accurately, and uniformly marked on the scale with a black pigment of a permanent and indelible character. The scale shall be graduated in 2-degree units. The 10-degree marks shall be longer than the other marks and shall be figured on the scale.

3.2.1.5 Guards.-

3.2.1.5.1 Class 2 wire mesh guards.- The tube and bulb of the thermometer shall be protected against external damage with wire cloth. The wire cloth shall conform to type I of Specification RR-C-440. The wire cloth shall be square mesh, plain weave, made of low carbon steel, hot or electro-galvanized after weaving; and shall have a 4 by 4 mesh with a wire diameter of 0.0348 inch. The wire cloth shall be bent over the case flanges for the full length of the flanges and crimped over the edges to secure the cloth to the flanges.

3.2.1.5.2 Class 3 plate guards.- The bulb of the thermometer shall be protected against external damage with a solid, slotted, or perforated metal plate attached to the lower end of the case. The metal plate may be a separate enclosure. The plate shall not prevent free circulation of air around the bulb.

3.2.2 Type II, shiphold suspension, air or water.-

3.2.2.1 Construction.- The tube shall be mounted on a corrosion-resistant metal scale having a black enameled face and the scale shall be mounted on a case of a design similar to figure 2. The case shall be of heavy galvanized cast iron approximately 1-5/16 inches in diameter and 15-1/4 inches long. It shall have a ring or hole in the top for suspension, and shall have a rubber plug in the bottom and rubber rings at each end to protect the instrument from injury in handling.

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3.2.2.2 Attachment.- The tube shall be mounted to the scale by means of metal clamps secured by eyelets and the hook on the tube shall fit into a hole in the scale to prevent vertical slippage. The scale back shall be screwed to the case and shall be adjusted to leave a free circulation of water around the bulb and shall be so positioned that the bulb is well protected by the case against breakage.

3.2.2.3 Graduations.- The graduation marks shall be neatly and accurately engraved into the scale back and the range numerals shall be die stamped into the scale for permanence. The graduation marks and figures shall be filled with white baking enamel.

3.3 Marking.- The name of the manufacturer shall be marked on the scale in accordance with Standard MIL-STD-130.

3.4 Interchangeability.- For like equipment, all removable units and replacement parts shall be interchangeable with corresponding items or parts without requiring hand or machine fitting.

3.5 Workmanship.- Workmanship shall be of the highest grade throughout and in accordance with good commercial practice for this type of equipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling.-

4.2.1 Lot.- For purpose of sampling, a lot shall consist of all thermometers of the same type offered for delivery at one time, except that, in no case shall a lot exceed 5,000 thermometers.

4.2.2 Sampling for examination and calibration and drainage tests.- Sample thermometers shall be selected at random in accordance with Standard MIL-STD-105 at Inspection Level I for the examination and tests specified in 4.3, 4.4.1, and 4.4.2. If any thermometer has one or more defects, or fails in either test, it shall be rejected. Lots shall be accepted or rejected in accordance with Acceptable Quality Level (AQL) 2.5 percent for each test separately.

4.2.3 Sampling for color stability test.- Sample thermometers shall be selected in accordance with Inspection Level L-4 of Standard MIL-STD-105 for the test specified in 4.4.3. If any thermometer fails in this test, the entire lot shall be rejected.

4.3 Examination.- Sample thermometers selected in accordance with 4.2.2 shall be examined for defects as defined in Standard MIL-STD-105 using an AQL of 2.5 percent defective for major defects and 6.5 percent defective for minor defects.

4.4 Test procedures.-

4.4.1 Calibration test.- The calibration of each of the sample thermometers selected in accordance with 4.2.2 shall be carefully checked to determine conformance with the accuracy specified in 3.1.4. Sample thermometers shall be tested in a well-stirred bath, compared with a thermometer certified by the National Bureau of Standards and under full immersion. Thermometers shall be tested at the top and bottom temperature points of the minimum range (minus 20 and plus 120 degrees) and at a point halfway between.

4.4.2 Drainage test.- Sample thermometers selected in accordance with 4.2.2 shall be subjected to a sudden temperature change of approximately 70° F. by quickly transferring from a water bath of approximately 100° F. to a crushed ice and water bath. If bubbles form in the bulb or column at any time during this test or if the indicating column is separated after 2 minutes immersion in the crushed ice bath, the thermometer shall be considered defective.

4.4.3 Color stability test.- Sample thermometers selected in accordance with 4.2.3 shall be subjected to the following color stability test. The liquid column shall be free from distillation and shall still be easily readable after 100 hours exposure in an Atlas Fade-ometer. This is equivalent to four months of continuous exposure to noontime June sunlight.

4.5 Defective units.- Units found defective in the course of inspection shall be corrected to the satisfaction of the Government or shall be replaced with nondefective units.

4.6 Inspection of preparation for delivery.- The preservation, packaging, packing, and marking of the boxes shall be inspected to determine compliance with the requirements of section 5 herein, and as required by specifications referenced herein. Except as required by referenced specifications, inspection lots and sampling shall be in accordance with Standard MIL-STD-105 with Inspection Level L7, and an AQL of 4.0 percent defective.

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5. PREPARATION FOR DELIVERY

5.1 Packaging (see 6.2).-

5.1.1 Level A.-

5.1.1.1 Unit packaging.- The thermometers shall be individually wrapped in not less than two sheets of tissue, and adequately cushioned with nonabrasive stuffing material, or protected with interior fiberboard packing in such a manner that clearance shall be maintained between the thermometer and all inside surfaces of the set-up, metal-stayed, or folding paperboard box into which it shall be individually packaged. The boxes shall conform to Specifications PPP-B-676, PPP-B-665, or PPP-B-566, respectively. Box closure shall be as specified in the applicable box specification or appendix thereto.

5.1.1.2 Intermediate packaging.- Thermometers shall be intermediate packaged in quantities of not more than twenty to a fiber box conforming to Specification PPP-B-636 (class 2). Box closure shall be as specified in the appendix to the box specification. All corners and edge seams and manufacturers joints shall be sealed with minimum 1-1/2 inch wide tape conforming to Specification PPP-T-76.

5.1.2 Level C.- The thermometers shall be packaged in accordance with the manufacturers commercial practice.

5.2 Packing (see 6.2).-

5.2.1 Level A.- Thermometers, packaged as specified shall be packed in overseas type, wood cleated fiberboard, nailed wood, fiber, wirebound wood, wood cleated veneer paper overlaid, or wood cleated plywood boxes conforming to Specification PPP-B-591, PPP-B-621, PPP-B-636 class 3, PPP-B-585, MIL-B-10377, or PPP-B-601, respectively at the option of the contractor. Boxes shall be closed and strapped in accordance with the applicable box specification or appendix thereto, except fiber boxes shall be banded with tape conforming to type IV of Specification PPP-T-97 and the appendix thereto. The gross weight of wood or wood cleated boxes shall not exceed 200 pounds; fiber boxes shall not exceed the weight limitations of the applicable box specification. Intermediate fiber boxes closed, sealed and banded as specified herein, and used as the shipping container need not be overpacked.

5.2.2 Level B.- Thermometers, packaged as specified shall be packed in domestic type wood cleated fiberboard, nailed wood, wirebound wood, cleated plywood or wood cleated veneer paper overlaid boxes or class 2 fiber boxes conforming to Specification PPP-B-591, PPP-B-621, PPP-B-585, PPP-B-601, MIL-B-10377 or PPP-B-636, respectively at the option of the contractor. Box closure shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood or wood cleated boxes shall not exceed 200 pounds; fiber boxes shall not exceed the weight limitations of the applicable box specification. Intermediate fiber boxes closed as specified herein, and used as the shipping container need not be overpacked.

5.2.3 Level C.- Thermometers, packaged as specified shall be packed in containers which will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply to the Uniform Freight Classification Rules or other regulations as applicable to the mode of transportation.

5.3 Marking.- In addition to any special marking required by the contract or order or herein, interior and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intended use.- The general purpose air thermometers covered by this specification are intended for use in determining the temperature of the surrounding air, for outdoor and indoor use and the suspension type thermometers are intended to determine air or water temperatures primarily in ship holds. These thermometers are not to be used for meteorological purposes.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and class required (see 1.2).
(c) When treatment and painting other than that specified is required (see 3.2.1.2).
(d) Level of packaging and level of packing required (see 5.1 and 5.2).

6.3 Bids for furnishing thermometers differing from this specification will be considered, provided the bid clearly describes the specific points in which the thermometers proposed differ from this specification and provided, further, that the differences are indicated as such. When exceptions are not clearly described and indicated as such, it will be assumed that bidders are offering thermometers in strict accordance with this specification.

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States

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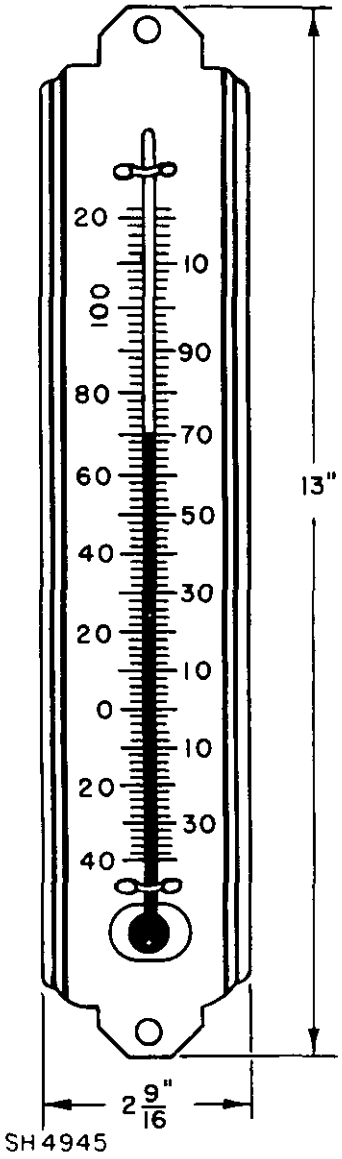


Figure 1 - Type I, general purpose, air

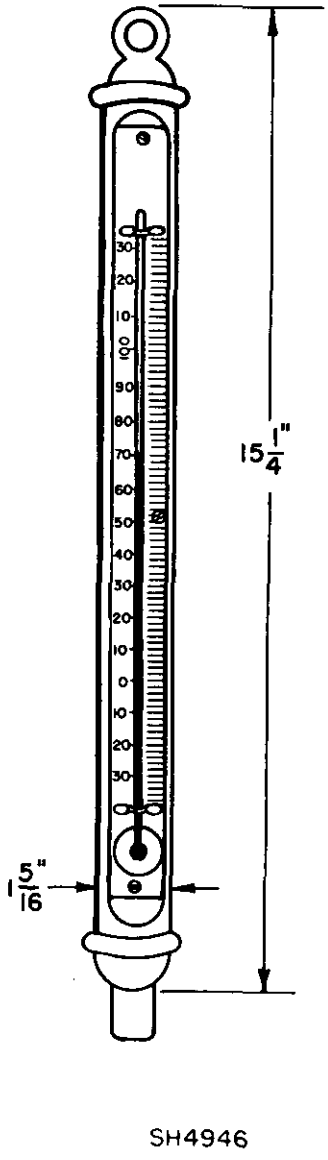


Figure 2 - Type II, shiphold suspension, air or water

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